COORDINATE GEOMETRY

Straight LIne

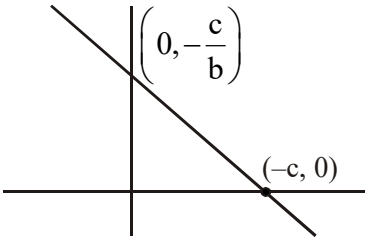
1. Let the area of the triangle formed by a straight line L : x + by + c = 0 with co-ordinate axes be 48 square units. If the perpendicular drawn from the origin to the line L makes an angle of 45° with the positive x-axis, then the value of $b^2 + c^2$ is :

माना एक सरल रेखा L : x + by + c = 0 और निर्देशांक अक्षों से बनने वाले त्रिभुज का क्षेत्रफल 48 वर्ग इकाई है। यदि रेखा L पर मूल बिन्दु से डाला गया लंब धनात्मक x-अक्ष से 45° का कोण बनाता है, तो $b^2 + c^2$ का मान है :

(A) 83 (B) 97 (C) 90 (D) 93

Ans. B

Sol.



$L : x + by + c = 0$

$\because \frac{1}{2}\left|(-c)\cdot\left(\frac{-c}{b}\right)\right| = 48$

$\therefore \left|\frac{c^2}{b}\right| = 96$ ...(1)

Slope of line $L = -\frac{1}{b}$

$\therefore$ Slope of line perpendicular to L is b.

$\therefore b = 1$

$\therefore c^2 = 96$

$\therefore b^2 + c^2 = 97$

2. A line passes through the origin and makes equal angles with the positive coordinate axes. It intersects the lines $L_1 : 2x + y + 6 = 0$ and $L_2 : 4x + 2y - p = 0$, $p > 0$, at the points A and B, respectively. If $AB = \frac{9}{\sqrt{2}}$ and the foot of the perpendicular from the point A on the line L_2 is M, then $\frac{AM}{BM}$ is equal to

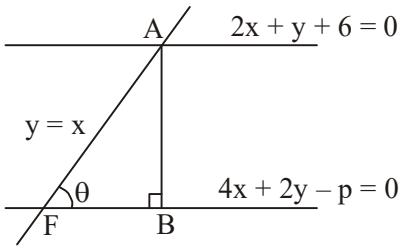
एक रेखा मूलबिन्दु से होकर जाती है और धनात्मक निर्देशांक अक्षों से बराबर कोण बनाती है। यह रेखाओं $L_1 : 2x + y + 6 = 0$ और

$L_2 : 4x + 2y - p = 0, p > 0$, को क्रमशः बिन्दुओं A और B पर काटती है। यदि $AB = \frac{9}{\sqrt{2}}$ है और बिन्दु A से रेखा L_2 पर लम्ब का पाद M है, तो $\frac{AM}{BM}$ बराबर है :

(A) 3 (B) 5 (C) 2 (D) 4

Ans. A

Sol.



$m_1 = 1$ $m_2 = -2$

$\tan\theta = \left|\frac{1+2}{1-2}\right| = 3$

$\tan\theta = \frac{AF}{BF} = 3$

3. Let the equation $x(x+2)(12-k) = 2$ have equal roots. Then the distance of the point $\left(k, \frac{k}{2}\right)$ from the line $3x + 4y + 5 = 0$ is :

माना समीकरण $x(x + 2)(12 - k) = 2$ के मूल बराबर हैं। तो बिन्दु $\left(k, \frac{k}{2}\right)$ की रेखा $3x + 4y + 5 = 0$ से दूरी है :

(A) $15\sqrt{5}$ (B) $5\sqrt{3}$ (C) 15 (D) 12

Ans. C

Sol. $(x^2 + 2x)(12 - k) = 2$

$\lambda x^2 + 2\lambda x - 2 = 0$ $k \neq 12$ Let $12 - k = \lambda$

$D = 0$

$4\lambda^2 + 8\lambda = 0$

$\lambda = 0$ or $\lambda = -2$

$\Rightarrow 12 - k = -2$

$k = 14$

So, $P\left(k, \frac{k}{2}\right) = (14, 7)$

$$d = \left|\frac{3\times14+4\times7+5}{5}\right| = 15$$

option (3)

4. Consider the lines $x(3\lambda + 1) + y(7\lambda + 2) = 17\lambda + 5$, λ being a parameter, all passing through a point P. One of these lines (say L) is farthest from the origin. If the distance of L from the point (3, 6) is d, then the value of d^2 is :

एक बिन्दु P से होकर जाने वाली रेखाओं $x(3\lambda + 1) + y(7\lambda + 2) = 17\lambda + 5$, जहाँ λ एक प्राचल है, का विचार कीजिए। इनमें से एक रेखा (माना L), मूल बिन्दु से अधिकतम दूरी पर है। यदि रेखा L की बिन्दु (3, 6) से दूरी d है, तो d^2 बराबर है :

(A) 15 (B) 30 (C) 20 (D) 10

Ans. C

Sol. $(x + 2y - 5) + \lambda(3x + 7y - 17) = 0$

$\Rightarrow P(1, 2)\ Q(3,6)$

$d^2 = PQ^2 = 4 + 16 = 20$

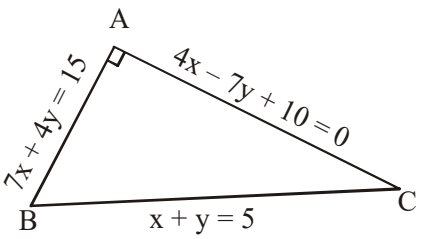
5. Let the three sides of a triangle are on the lines $4x - 7y + 10 = 0$, $x + y = 5$ and $7x + 4y = 15$. Then the distance of its orthocentre from the orthocentre of the tringle formed by the lines $x = 0$, $y = 0$ and $x + y = 1$ is :

माना एक त्रिभुज की तीन भुजाऐं रेखाओं $4x - 7y + 10 = 0$, $x + y = 5$ और $7x + 4y = 15$ पर है। तो इसके लंबकेन्द्र की, रेखाओं $x = 0$, $y = 0$ और $x + y = 1$ से बने त्रिभुज के लंबकेन्द्र से दूरी है :

(A) $\sqrt{5}$ (B) 5 (C) $\sqrt{20}$ (D) 20

Ans. A

Sol.



As $7x + 4y = 15 \perp$ to $4x - 7y + 10 = 0$

$\therefore \angle A$ is 90°

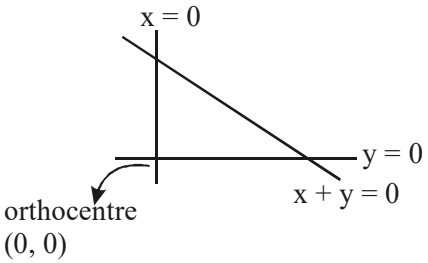
In case of Right Angled Δ, orthocentre is vertex on which 90° is formed.

$4x - 7y + 10 = 0$

& $7x + 4y - 15 = 0$

$x = 1, y = 2$

Again



Distance = $\sqrt{1^2+2^2}$

$=\sqrt{5}$

6. For an integer $n \geq 2$, if the arithmetic mean of all coefficients in the binomial expansion of $(x+y)^{2n-3}$ is 16 , then the distance of the point $P(2n-1, n^2-4n)$ from the line $x+y=8$ is :

एक पूर्णांक $n \geq 2$ के लिए, यदि $(x+y)^{2n-3}$ के द्विपद प्रसार में सभी गुणांकों का समान्तर माध्य 16 है, तो बिन्दु $P(2n-1, n^2-4n)$ की रेखा $x+y=8$ से दूरी है :

(A) $\sqrt{2}$ (B) $3\sqrt{2}$ (C) $2\sqrt{2}$ (D) $5\sqrt{2}$

Ans. B

Sol. $n \geq 2$

$(x+y)^{2n-3}$

A.M. of all Binomial coefficient = 16

i.e. $\frac{\text{Sum of B.C.}}{\text{Total}} = 16$

$\frac{2^{2n-3}}{2n-2} = 16$

$2^{2n-3} = 2^4.2(n-1)$

$2^{2n-3} = 2^5(n-1)$

$2^{2n-8} = n-1$

$n = 5$ By Hit & Trial

$2^2 = 5-1$

$= 4$

$\therefore P(2n-1, n^2-4n)$

from $x+y-8=0$

P(9, 5) $d = \frac{|9+5-8|}{\sqrt{2}}$

$= \frac{6}{\sqrt{2}} = 3\sqrt{2}$

7. Let ABC be the triangle such that the equations of lines AB and AC be $3y-x=2$ and $x+y=2$, respectively, and

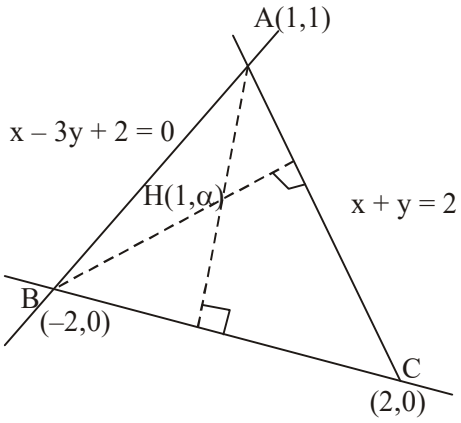
the points B and C lie on x-axis. If P is the orthocentre of the triangle ABC , then the area of the triangle PBC is equal to :

माना त्रिभुज ABC की भुजाओं AB और AC के समीकरण क्रमशः $3y - x = 2$ और $x + y = 2$ हैं और बिन्दु B तथा C, x-अक्ष पर हैं। यदि त्रिभुज ABC का लम्बकेन्द्र P है, तो त्रिभुज PBC का क्षेत्रफल बराबर है :

(A) 6 (B) 4 (C) 8 (D) 10

Ans. A

Sol.



$\frac{\alpha - 0}{1-(-2)} \times (-1) = -1 \Rightarrow \alpha = 3$ So H(1,3)

Area of triangle $BHC = \frac{1}{2} \times 4 \times 3 = 6$

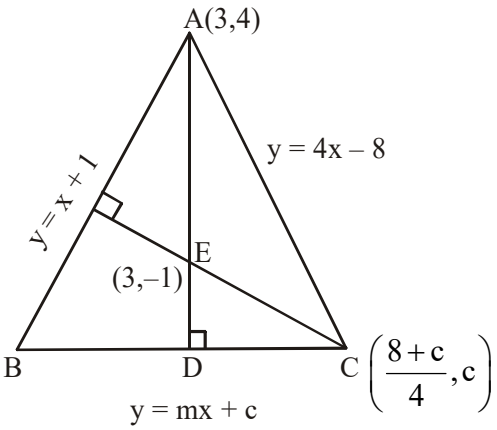
8. If the orthocenter of the triangle formed by the lines $y = x + 1$, $y = 4x - 8$ and $y = mx + c$ is at $(3, -1)$, then $m - c$ is :

यदि रेखाओं $y = x + 1$, $y = 4x - 8$ और $y = mx + c$ द्वारा बनाए गए त्रिभुज का लम्बकेन्द्र $(3, -1)$ है, तो $m - c$ बराबर है :

(A) –2 (B) 4 (C) 0 (D) 2

Ans. C

Sol.



$\because m_{AD} \to \infty$

so, $m_{BC} = 0$

$$m_{CE} = \frac{c+1}{\frac{8+c}{4}-3} = \frac{4c+4}{c-4}$$

$m_{AB} \cdot m_{CE} = -1$

$$\Rightarrow 1\cdot\left(\frac{4c+4}{c-4}\right) = -1$$

$\Rightarrow 4c + 4 = 4 - c$

$\Rightarrow c = 0$

So, $m = c = 0$

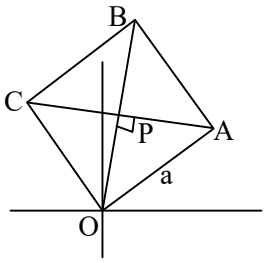
9. Let a be the length of a side of a square OABC with O being the origin. Its side OA makes an acute angle α with the positive x-axis and the equations of its diagonals are $\left(\sqrt{3}+1\right)x+\left(\sqrt{3}-1\right)y=0$ and $\left(\sqrt{3}-1\right)x-\left(\sqrt{3}+1\right)y+8\sqrt{3}=0$. Then a^2 is equal to :

माना वर्ग OABC की एक भुजा की लंबाई a है, जहाँ O मूल बिन्दु है। इसकी OA धनात्मक x-अक्ष से न्यून कोण α बनाती है और इसके विकर्णो के समीकरण $\left(\sqrt{3}+1\right)x+\left(\sqrt{3}-1\right)y=0$ तथा $\left(\sqrt{3}-1\right)x-\left(\sqrt{3}+1\right)y+8\sqrt{3}=0$ हैं। तो a^2 बराबर है :

(A) 16 (B) 24 (C) 48 (D) 32

Ans. C

Sol.



Point of intersection of diagonals

$P\left(-\sqrt{3}(\sqrt{3}-1), \sqrt{3}\left(\sqrt{3}+1\right)\right)$

$OP = \sqrt{24}$

Now, $OA^2 = OP^2 + PA^2$

$a^2 = 24 + 24$

$a^2 = 48$

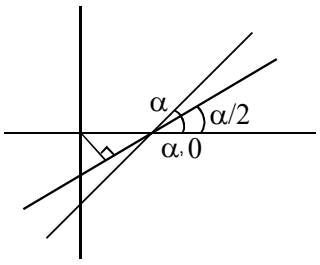
10. A line passing through the point P(a, 0) makes an acute angle α with the positive x-axis. Let this line be rotated about the point P through an angle $\frac{\alpha}{2}$ in the clock-wise direction. If in the new position, the slope of the line is $2-\sqrt{3}$ and its distance from the origin is $\frac{1}{\sqrt{2}}$, then the value of $3a^2 \tan^2 \alpha - 2\sqrt{3}$ is :

बिन्दु P(a, 0) से होकर जाने वाली एक रेखा धनात्मक x-अक्ष से न्यून कोण α बनाती है। माना इस रेखा को बिन्दु P के सापेक्ष घड़ी की दिशा में $\frac{\alpha}{2}$ कोण तक घुमाया गया है। यदि नई स्थिति में इस रेखा की प्रवणता $2-\sqrt{3}$ है और मूल बिन्दु से इसकी दूरी $\frac{1}{\sqrt{2}}$ है, तो $3a^2 \tan^2 \alpha - 2\sqrt{3}$ का मान है :

(A) 4 (B) 8 (C) 5 (D) 6

Ans. A

Sol.



$\tan\frac{\alpha}{2} = 2-\sqrt{3}$

$\alpha = 30°$

$y-0=\left(2-\sqrt{3}\right)(x-a)$

$\frac{1}{\sqrt{2}}=\frac{\left|\left(2-\sqrt{3}\right)a\right|}{\sqrt{4+3-4\sqrt{3}+1}}$ $\quad |a|\left(2-\sqrt{3}\right)=\frac{2}{\sqrt{2}}\sqrt{2-\sqrt{3}}$ $\quad \Rightarrow a=\frac{\sqrt{2}}{\sqrt{2-\sqrt{3}}}$

$3a^2\tan^2\alpha-2\sqrt{3}=3\times2\left(2+\sqrt{3}\right)\frac{1}{3}-2\sqrt{3}=4$

11. Let the triangle PQR be the image of the triangle with vertices (1, 3),(3, 1) and (2, 4) in the line $x+2y=2$. If the centroid of ΔPQR is the point (α, β), then $15(\alpha-\beta)$ is equal to :

माना कि त्रिभुज PQR, रेखा $x+2y=2$ में शीर्षों (1, 3),(3, 1) और (2, 4) के साथ एक त्रिभुज का प्रतिबिम्ब है। यदि ΔPQR का केन्द्रक, बिन्दु (α, β) है, तो $15(\alpha-\beta)$ बराबर है :

(A) 19 (B) 21 (C) 24 (D) 22

Ans. D

Sol. Let 'G' be the centroid of Δ formed by (1, 3) (3, 1) & (2, 4)

$G\cong\left(2,\frac{8}{3}\right)$

Image of G w.r.t. $x+2y-2=0$

$\frac{\alpha-2}{1}=\frac{\beta-\frac{8}{3}}{2}=-2\frac{\left(2+\frac{16}{3}-2\right)}{1+4}$

$=\frac{-2}{5}\left(\frac{16}{3}\right)$

$\Rightarrow \alpha=\frac{-32}{15}+2=\frac{-2}{15}, \beta=\frac{-32\times2}{15}+\frac{8}{3}=\frac{-24}{15}$

$15(\alpha-\beta)=-2+24=22$

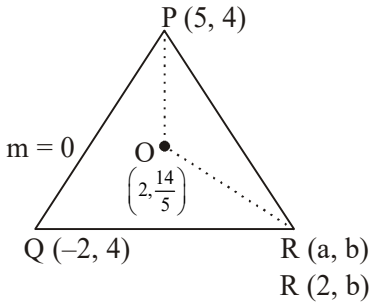
12. Let the area of a ΔPQR with vertices P(5,4), Q(–2,4) and R(a, b) be 35 square units. If its orthocenter and centroid are $O\left(2,\frac{14}{5}\right)$ and C(c, d) respectively, then $c+2d$ is equal to :

माना कि शीर्षों P(5,4), Q(–2,4) और R(a, b) के साथ एक ΔPQR का क्षेत्रफल 35 वर्ग इकाई है। यदि इसके लम्बकेन्द्र और केन्द्रक क्रमशः $O\left(2,\frac{14}{5}\right)$ और C(c, d) हैं तो $c+2d$ बराबर है :

(A) 2 (B) $\frac{8}{3}$ (C) $\frac{7}{3}$ (D) 3

Ans. D

Sol.



$$\begin{vmatrix} 5 & 4 & 1 \\ -2 & 4 & 1 \\ a & b & 1 \end{vmatrix} = 70$$

$R_1 \to R_1 - R_2$

$|b - 4| = 10$

$b = 14$ or $b = -6$

$m_{PQ} \cdot m_{OR} = -1$

$m_{OR} \to \infty$

$a = 2$

$$C\left(\frac{5}{3},\frac{2}{3}\right)$$

$$c + 2d = \frac{5}{3} + \frac{4}{3} = \frac{9}{3} = 3$$

$$\Rightarrow \begin{vmatrix} 7 & 0 & 0 \\ -2 & 4 & 1 \\ a & b & 1 \end{vmatrix} = 70$$

$|7(4 - b)| = 70$

$m_{OP} \cdot m_{QR} = -1$

$$\frac{4-\frac{14}{5}}{3} \times \frac{b-4}{4} = -1$$

$$\frac{2}{5} \times \frac{b-4}{4} = -1$$

$b - 4 = -10$

$b = -6$

13. A rod of length eight units moves such that its ends A and B always lie on the lines $x - y + 2 = 0$ and $y + 2 = 0$, respectively. If the locus of the point P, that divides the rod AB internally in the ratio 2 : 1 is $9(x^2 + \alpha y^2 + \beta xy + \gamma x + 28y) - 76 = 0$, then $\alpha - \beta - \gamma$ is equal to :

आठ इकाई लम्बी एण्क छड़ इस प्रकार चलायमान है कि इसके सिरे A और B सदैव क्रमशः रेखाओं $x - y + 2 = 0$ और $y + 2 = 0$ पर रहते हैं। यदि बिन्दु P, जो छड़ AB को अन्तः 2 : 1 के अनुपात में विभाजित करता है, का बिन्दुपथ $9(x^2 + \alpha y^2 + \beta xy + \gamma x + 28y) - 76 = 0$ है, तो $\alpha - \beta - \gamma$ बराबर है :

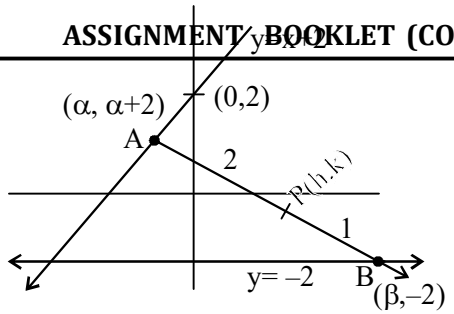
(A) 22 (B) 24 (C) 21 (D) 23

Ans. D

Sol. $y = x + 2$

$$h = \frac{2\beta + \alpha}{3}$$

$\Rightarrow 2\beta = 3h - \alpha$



$\& \ k = \frac{\alpha + 2 - 4}{3}$

$\Rightarrow k = \frac{\alpha - 2}{3}$

$\Rightarrow \alpha = 3k + 2$

$2\beta = 3h - (3k + 2)$

$2\beta = 3h - 3k - 2$

$|AB| = 8$

$(\alpha - \beta)^2 + (\alpha + 4)^2 = 64$

$$\left\{3k + 2 - \frac{3h - 3k - 2}{2}\right\}^2 + (3k + 2 + 4)^2 = 64$$

$9[(3k - h + 2)^2 + 4(k + 2)^2] = 64 \times 4$

$9[x^2 + 13y^2 - 6xy - 4x + 28y] = 76$

$\alpha - \beta - \gamma = 13 + 6 + 4 = 23$

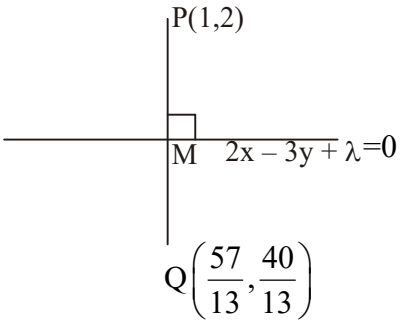
14. Let the lines $3x - 4y - \alpha = 0$, $8x - 11y - 33 = 0$, and $2x - 3y + \lambda = 0$ be concurrent. If the image of the point(1, 2) in the line $2x - 3y + \lambda = 0$ is $\left(\frac{57}{13}, \frac{-40}{13}\right)$, then $|\alpha\lambda|$ is equal to :

माना कि रेखाएँ $3x - 4y - \alpha = 0$, $8x - 11y - 33 = 0$, और $2x - 3y + \lambda = 0$ संगामी हैं। यदि रेखा $2x - 3y + \lambda = 0$ में बिन्दु (1, 2) का प्रतिबिम्ब $\left(\frac{57}{13}, \frac{-40}{13}\right)$ है, तो $|\alpha\lambda|$ बराबर है :

(A) 91 (B) 84 (C) 101 (D) 113

Ans. A

Sol.



$\because$ PM = QM

So, $M\left(\frac{\frac{57}{13}+1}{2}, \frac{\frac{-40}{13}+2}{2}\right)$

$=\left(\frac{35}{13}, \frac{-7}{13}\right)$

$\because$ M lies on the line.

$2x - 3y + \lambda = 0$

$2\left(\frac{35}{13}\right) - 3\left(\frac{-7}{13}\right) + \lambda = 0$

$\lambda = -\frac{70}{13} + \frac{21}{13}$

$= \frac{-91}{13} = -7$

$$\begin{vmatrix} 3 & -4 & -\alpha \\ 8 & -11 & -33 \\ 2 & 3 & \lambda \end{vmatrix} = 0$$

$\Rightarrow 3(-11\lambda - 99) + 4(8\lambda + 66) - \alpha(-24 + 22) = 0$
$\Rightarrow 33\lambda - 297 + 32\lambda + 264 + 24\alpha - 22\alpha = 0$
$\Rightarrow -\lambda + 2\alpha - 33 = 0$ ...(1)
$\therefore \lambda = -7$
$-(-7) + 2\alpha - 33 = 0$
$2\alpha = 26$
$\alpha = 13$
$\therefore |\alpha\lambda| = |13 \times (-7)|$
$= 91$

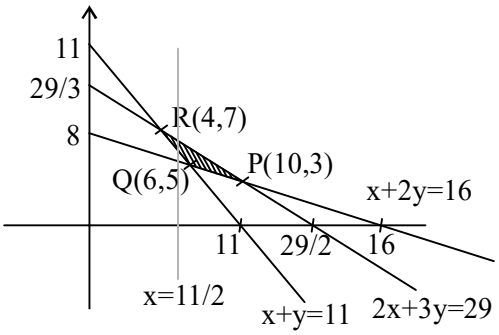
15. Let the points $\left(\frac{11}{2}, \alpha\right)$ lie on or inside the triangle with sides $x + y = 11$, $x + 2y = 16$ and $2x + 3y = 29$. Then the product of the smallest and the largest values of α is equal to :

माना कि बिन्दु $\left(\frac{11}{2}, \alpha\right)$ भुजाओं $x + y = 11$, $x + 2y = 16$ और $2x + 3y = 29$ के एक त्रिभुज पर या उसके अन्दर स्थित हैं। तब α के न्यूनतम और महत्तम मानों का गुणनफल बराबर है :

(A) 22 (B) 44 (C) 33 (D) 55

Ans. C

Sol.



for α_{min}, put $x = \frac{11}{2}$ in $x + y = 11$

$\alpha_{min} = \frac{11}{2}$

for α_{max}, put $x = \frac{11}{2}$ in $2x + 3y = 29$

$\alpha_{max} = 6$

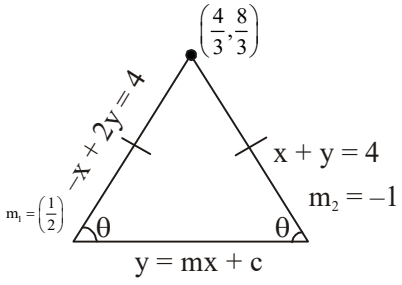
16. Two equal sides of an isosceles triangle are along $-x + 2y = 4$ and $x + y = 4$. If m is the slope of its third side, then the sum, of all possible distinct values of m, is :

एक समद्विबाहु त्रिभुज की दो बराबर भुजाएँ, $-x + 2y = 4$ और $x + y = 4$ के अनुदिश हैं। यदि m इसकी तीसरी भुजा की प्रवणता है, तो m के सभी संभव भिन्न मानों का योग है :

(A) $-2\sqrt{10}$ (B) 12 (C) 6 (D) -6

Ans. C

Sol.



$$\Rightarrow \left|\frac{m - m_1}{1 + m_1 m}\right| = \left|\frac{m - m_2}{1 + m m_2}\right|$$

$$\Rightarrow \left|\frac{m - \frac{1}{2}}{1 + \frac{m}{2}}\right| = \frac{|m+1|}{|1-m|}$$

$\Rightarrow \left|\frac{2m-1}{2+m}\right| = \left|\frac{1+m}{1-m}\right|$

$\Rightarrow (2m-1)(1-m) = \pm(2+m)(1+m)$

Case : 1 $\Rightarrow 2m - 2m^2 - 1 + m = 2 + 2m + m + m^2$

$\Rightarrow -3 = 3m^2 \rightarrow$ Not possible

Case : 2 $2m - 2m^2 - 1 + m = -2 - 2m - m - m^2$

$m^2 - 6m - 1 = 0$

$m_1 + m_2 = 6$

17. Let ABC be a triangle formed by the lines $7x - 6y + 3 = 0$, $x + 2y - 31 = 0$ and $9x - 2y - 19 = 0$. Let the point (h, k) be the image of the centroid of ΔABC in the line $3x + 6y - 53 = 0$. Then $h^2 + k^2 + hk$ is equal to:

माना कि ABC, रेखाओं $7x - 6y + 3 = 0$, $x + 2y - 31 = 0$ और $9x - 2y - 19 = 0$ द्वारा निर्मित एक त्रिभुज है। माना कि बिन्दु (h, k), ΔABC के केन्द्रक का रेखा $3x + 6y - 53 = 0$ में प्रतिबिम्ब है। तब $h^2 + k^2 + hk$ बराबर है :

(A) 47 (B) 37 (C) 40 (D) 36

Ans. B

Sol. $7x - 6y + 3 = 0$

$x + 2y - 31 = 0$

$7x - 6y = -3$

$7x + 14y = 217$

$-20y = -220$

$y = 11$

$x + 22 - 31 = 0$

$x = 9$

(9, 11)

$x + 2y = 31$

$9x - 2y = 19$

$9x + 18y = 279$

$9x - 2y = 19$

$20y = 260$

$y = 13$

$x + 26 = 31$

$x = 5$

(5, 13)

$9x - 2y = 19$

$7x - 6y = -3$

$63x - 14y = 133$

$63x - 54y = -27$

$40y = 160$

$y = 4$

$9x - 8 = 19$

$9x = 27$

$x = 3$

(3, 4)

$G = \left(\frac{17}{3}, \frac{28}{3}\right)$

$$\frac{x - \frac{17}{3}}{3} = \frac{y - \frac{28}{3}}{6} = -2\left(\frac{17 + 56 - 53}{45}\right)$$

$$\frac{x - \frac{17}{3}}{3} = \frac{y - \frac{28}{3}}{6} = -2\left(\frac{20}{45}\right)$$

$$\frac{x - \frac{17}{3}}{3} = \frac{y - \frac{28}{3}}{6} = \frac{-8}{9}$$

$$x - \frac{17}{3} = \frac{y - \frac{28}{3}}{2} = -\frac{8}{3}$$

$$x = \frac{17}{3} - \frac{8}{3} \qquad y - \frac{28}{3} = -\frac{16}{3}$$

$$x = 3 \qquad y = \frac{12}{3} = 4$$

(3, 4)

$h = 3, k = 4$

$h^2 + k^2 + hk = 9 + 16 + 12 = 37$

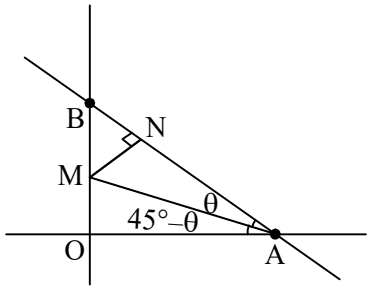
18. Let the line $x + y = 1$ meet the axes of x and y at A and B, respectively. A right angled triangle AMN is inscribed in the triangle OAB, where O is the origin and the points M and N lie on the lines OB and AB respectively. If the area of the triangle AMN is $\frac{4}{9}$ of the area of the triangle OAB and AN : NB = λ : 1, then the sum of all possible value(s) of is λ :

माना कि रेखा $x + y = 1$, x और y अक्षों को क्रमशः A और B पर मिलती है। एक समकोण त्रिभुज AMN, त्रिभुज OAB के अंतर्गत हैं, जहाँ O मूल–बिन्दु है और बिन्दु M और N क्रमशः रेखाओं OB और AB पर हैं। यदि त्रिभुज AMN को क्षेत्रफल, त्रिभुज OAB के क्षेत्रफल का $\frac{4}{9}$ है और AN : NB = λ : 1 है, तो λ के संभव मान/मानों का योग है :

(A) 1/2 (B) 5/2 (C) 13/6 (D) 2

Ans. D

Sol.



Area of $\Delta AOB = \frac{1}{2}$

Area of $\Delta AMN = \frac{4}{9} \times \frac{1}{2} = \frac{2}{9}$

Equation of AB is $x + y = 1$

$OA = 1$, $AM = \sec(45° - \theta)$

$AN = \sec(45° - \theta) \cos\theta$

$MN = \sec(45° - \theta) \sin\theta$

$Ar(\Delta AMN) = \frac{1}{2} \times \sec^2(45° - \theta) \sin\theta . \cos\theta = \frac{2}{9}$

$\Rightarrow \tan\theta = 2, \frac{1}{2}$

$\tan\theta = 2$ is rejected

$\frac{AN}{NB} = \frac{\lambda}{1} = \cot\theta = 2$

19. The vertices of a triangle are A(–1,3), B(–2,2) and C(3, –1). A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is :

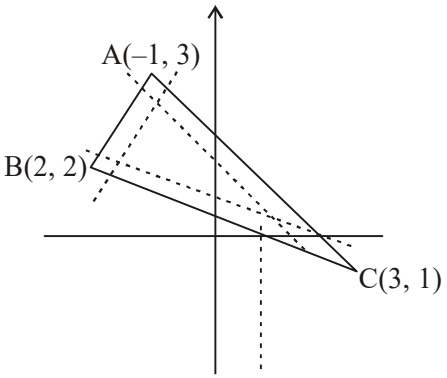
एक त्रिभुज के शीर्ष A(–1,3), B(–2,2) तथा C(3, –1) हैं। इस त्रिभुज की भुजाओं को त्रिभुज के अंदर की तरफ एक इकाई स्थानांतरित कर एक नया त्रिभुज बनाया गया है। तो नए त्रिभुज की मूलबिन्दु के सबसे निकट भुजा का समीकरण है :

(A) $x+y-\left(2-\sqrt{2}\right)=0$ (B) $x-y-\left(2+\sqrt{2}\right)=0$

(C) $-x+y-\left(2-\sqrt{2}\right)=0$ (D) $x+y+\left(2-\sqrt{2}\right)=0$

Ans. A

Sol.



$AB: y-2=\frac{1}{1}(x+2)$

$x-y-4=0$. $x-y+c=0$

$\left|\frac{c+4}{\sqrt{2}}\right|=1$

$C=\sqrt{2}-4,-\sqrt{2}-4$

$BC:\ y+1=\frac{3}{-5}(x-3)$

$-5y-5=3x-9$

$3x+5y=4$

$\left|\frac{c+4}{\sqrt{34}}\right|=1$

$AC: y-3=-1(x+1)$

$x+y-2$

$\left|\frac{c+2}{\sqrt{2}}\right|=1$

$C = \sqrt{2}-2, -\sqrt{2}-2$

Nearest : $x + y - 2 + \sqrt{2} = 0$

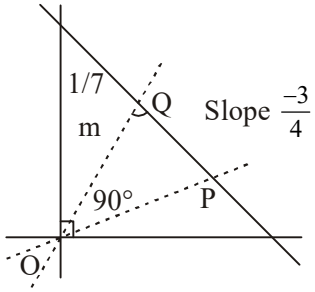
20. Let two straight lines drawn from the origin O intersect the line $3x + 4y = 12$ at the points P and Q such that ΔOPQ is an isosceles triangle and $\angle POQ = 90^\circ$. If $l = OP^2 + PQ^2 + QO^2$, then the greatest integer less than or equal to l is :

माना मूलबिन्दु से खींची गई दो रेखाएँ, रेखा $3x + 4y = 12$ को बिन्दुओं P तथा Q पर इस प्रकार काटती है कि ΔOPQ एक समद्विबाहु त्रिभुज है तथा $\angle POQ = 90^\circ$ है। यदि $l = OP^2 + PQ^2 + QO^2$ है, तो l से कम या l के बराबर महत्तम पूर्णांक है :

(A) 44 (B) 42 (C) 46 (D) 48

Ans. C

Sol. $\left|\dfrac{m+\frac{3}{4}}{1-\frac{3m}{4}}\right| = 1$



$m + \dfrac{3}{4} = 1 - \dfrac{3}{4}m$

$m + \dfrac{3}{4}m = 1 - \dfrac{3}{4}$

$7m = 1$

$\boxed{m = \dfrac{1}{7}}$

$OQ = y = \dfrac{1}{7}x$ $\qquad$ OP $y = 7x$

$Q = 3x + 4y = 12$

$3x + \dfrac{4}{7}x = 12$

$x = \dfrac{12\times 7}{25}, y = \dfrac{12\times 7}{25\times 7} = \dfrac{12}{25}, Q\left(\dfrac{84}{25}, \dfrac{12}{25}\right)$

$P \Rightarrow -25x = 12$

$x = \dfrac{-12}{25}, y = \dfrac{84}{25}$

$$P=\left(\frac{-12}{25},\frac{84}{25}\right)$$

$$OQ=\sqrt{\frac{84^2}{25^2}+\frac{12^2}{25^2}}$$

$$OP=\sqrt{\frac{84^2}{25^2}+\frac{12^2}{25^2}}$$

$$PQ=\sqrt{2}\sqrt{\frac{84^2}{25^2}+\frac{12^2}{25^2}}$$

$\ell = OQ^2 + OP^2 + PQ^2$

$$=\frac{84^2}{25^2}\times 2+\frac{12^2}{25^2}\times 2+2\left(\frac{84^2}{25^2}+\frac{12^2}{25^2}\right)$$

$$=4\times\frac{84^2}{25^2}+4\times\frac{12^2}{25^2}$$

$$\ell=4\times\frac{12^2}{25^2}\left(7^2+1\right)=\frac{144\times 8}{25}=46.08$$

$[\ell]=46$

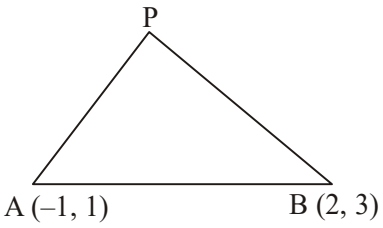
21. Let A (–1, 1) and B(2, 3) be two points and P be a variable point above the line AB such that the area of ΔPAB is 10. If the locus of P is ax + by = 15, then 5a + 2b is :

माना दो बिन्दु A (–1, 1) तथा B(2, 3) हैं और रेखा AB के ऊपर एक चर बिन्दु P इस प्रकार है कि ΔPAB का क्षेत्रफल 10 है। यदि P का बिंदुपथ ax + by = 15 है, तो 5a + 2b बराबर है :

(A) 6 (B) 4 (C) $-\frac{6}{5}$ (D) $-\frac{12}{5}$

Ans. D

Sol.



Locus of P,
ax + by = 15

Take point P as (h, k)

Area of Δ ABP

$= \frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) | = 10$

$= \frac{1}{2} | -1(3-k) + 2(k-1) + h(1-3) | = 10$

$= | -3 + k + 2k - 2 + h - 3h | = 20$

$= | 3k - 2h - 5 | = 20$

$= -(3k - 2h - 5) = 20 \Rightarrow 2h - 3k = 15$

$a = 2, b = -3$

$\therefore\ 5a + 2b = 4$

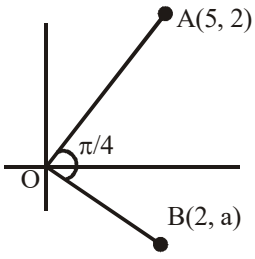
22. If the line segment joining the points (5,2) and (2,a) subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is :

यदि बिन्दुओं (5,2) तथा (2,a) को मिलाने वाला रेखाखंड, मूलबिन्दु पर $\frac{\pi}{4}$ का कोण बनाता है, तो a के सभी संभव मानों के गुणनफल का निरपेक्ष मान है :

(A) 2 (B) 8 (C) 4 (D) 6

Ans. C

Sol.



$M_{OA} = \frac{2}{5}$

$M_{OB} = \frac{a}{2}$

$= \tan\frac{\pi}{4} = \left| \frac{\frac{2}{5} - \frac{a}{2}}{1 + \frac{2}{5}\cdot\frac{a}{2}} \right|$

$\Rightarrow 1 = \left| \frac{4 - 5a}{10 + 2a} \right| \Rightarrow a = \frac{-6}{7}, \frac{14}{3}$

Product $\Rightarrow -4$, Abs. value = 4

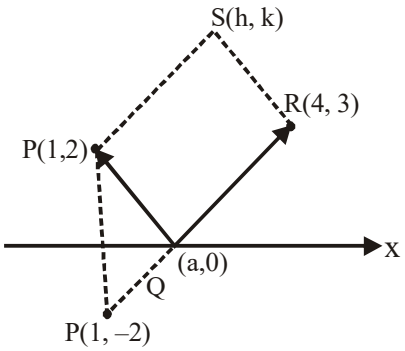
23. A ray of light coming from the point P(1, 2) gets reflected from the point Q on the x-axis and then passes through the point R(4, 3). If the point S(h, k) is such that PQRS is a parallelogram, then hk^2 is equal to :

बिन्दु P(1, 2) से आने वाली रोशनी की एक किरण x-अक्ष पर बिन्दु Q से परावर्तित होने के पश्चात् बिन्दु R(4, 3) से गुजरती है। यदि बिन्दु S(h, k) इस प्रकार है कि PQRS एक समांतर चतुर्भुज है, तो hk^2 बराबर है :

(A) 80 (B) 90 (C) 60 (D) 70

Ans. D

Sol.



$$P'R : y+2=\frac{5}{3}(x-1)$$

For Point $Q \Rightarrow y=0$

$$\frac{6}{5}=a-1 \Rightarrow a=\frac{11}{5}$$

Now, PQRS is parallelogram

$$\therefore \frac{h+a}{2}=\frac{4+1}{2} \Rightarrow h=5-\frac{11}{5}=\frac{14}{5}$$

and $\frac{2+3}{2}=\frac{k}{2} \Rightarrow K=5$

Now $hk^2 = 25\times\frac{14}{5}=14\times5=70$

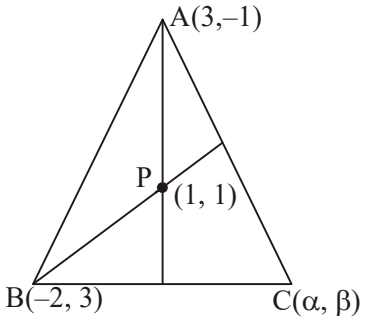
24. Two vertices of a triangle ABC are A(3,–1) and B(–2,3), and its orthocentre is P(1,1). If the coordinates of the point C are (α, β) and the centre of the circle circumscribing the triangle PAB is (h, k), then the value of $(\alpha+\beta)+2(h+k)$ equals :

किसी त्रिभुज ABC के दो शीर्ष A(3,–1) और B(–2,3) हैं तथा उसका लंब केन्द्र P(1,1) है। यदि बिन्दु C के निर्देशांक (α, β) हैं तथा त्रिभुज PAB के परिवृत्त का केन्द्र (h, k) है, तब $(\alpha+\beta)+2(h+k)$ का मान बराबर है :

(A) 5 (B) 51 (C) 15 (D) 81

Ans. A

Sol.



Equation of PC : $y-1=\frac{5}{4}(x-1)\ \left(\because m_{AB}=-\frac{4}{5}\right)$

$4y - 4 = 5x - 5$

$5x - 4y = 1$ _____(1)

Equation of BC : $y - 3 = 1(x + 2)\ (\because m_{AP} = -1)$

$x - y = -5$ _____(2)

Solving equation (1) and (2), $x = 21, y = 26$

$\therefore\ \alpha = 21, \beta = 26 \Rightarrow (\alpha + \beta) = 47$

Now, equation of $\perp$ bisector of AP : $y - 0 = 1(x - 2)$

$x - y = 2$ _____(3)

Equation of $\perp$ bisector of AB : $y-1=\frac{5}{4}\left(x-\frac{1}{2}\right)$ _____(4)

Solving equation (3) and (4),

$x=-\frac{19}{2}, y=-\frac{23}{2}$

i.e $2h = -19, 2k = -23$

$\therefore\ 2(h + k) = -42$

So, $(\alpha + \beta) + 2(h + k) = 47 - 42 = 5$

25. The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is :

रेखा $4x + 5y = 20$ के प्रथम चतुर्थांश में भाग को मूल बिन्दु से होकर जाने वाली रेखाएँ L_1 तथा L_2 समत्रिभाजित करती हैं। L_1 तथा L_2 के बीच एक कोण की स्पर्शज्या है :

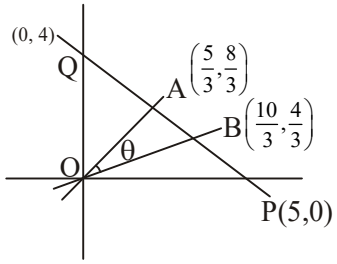
(A) $\frac{8}{5}$ (B) $\frac{2}{5}$ (C) $\frac{30}{41}$ (D) $\frac{25}{41}$

Ans. C

Sol. $A\left(\frac{5}{3},\frac{8}{3}\right)$, $B\left(\frac{10}{3},\frac{4}{3}\right)$

$m_{OA} = m_2 = \frac{8}{5}$

$m_{0B} = m_1 = \frac{2}{5}$



$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{\frac{8}{5} - \frac{2}{5}}{1 + \frac{8}{5}\times\frac{2}{5}}\right|$$

$$= \frac{\frac{6}{5}}{1 + \frac{16}{25}} = \frac{6\times 5}{25 + 16} = \frac{30}{41}$$

26. Let R be the interior region between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a, for which the points $\left(a^2, a+1\right)$ lie in R, is :

माना रेखाओं $3x - y + 1 = 0$ तथा $x + 2y - 5 = 0$ के अंदर का क्षेत्र, जिसमें मूलबिन्दु है, R है। a के सभी मानों, जिनके लिए $\left(a^2, a+1\right)$ क्षेत्र R में है, का समुच्चय है :

(A) $(-3,-1)\cup\left(-\frac{1}{3},1\right)$ (B) $(-3,-1)\cup\left(\frac{1}{3},1\right)$

(C) $(-3,0)\cup\left(\frac{1}{3},1\right)$ (D) $(-3,0)\cup\left(\frac{2}{3},1\right)$

Ans. C

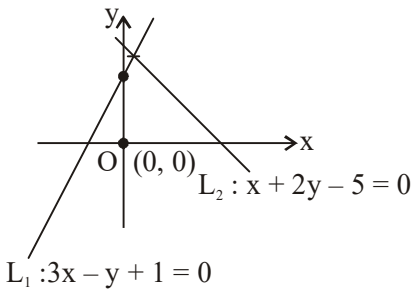
Sol. $P(a^2, a + 1)$

$L_1 = 3x - y + 1 = 0$

Origin and P lies same side w.r.t. L_1

$\Rightarrow L_1(0) \cdot L_1(P) > 0$

$\therefore 3(a^2) - (a + 1) + 1 > 0$



$\Rightarrow 3a^2 - a > 0$

$a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$ _____(1)

Let $L_2 : x + 2y - 5 = 0$

Originand P lies same side w.r.t. L_2

$\Rightarrow L_2(0) \cdot L_2(P) > 0$

$\Rightarrow a^2 + 2(a + 1) - 5 < 0$

$\Rightarrow a^2 + 2a - 3 < 0$

$\Rightarrow (a + 3)(a - 1) < 0$

$\therefore a \in (-3, 1)$ _____(2)

Intersection of (1) and (2)

$a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

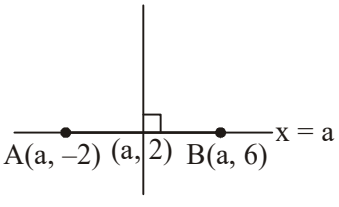
27. Let $\left(5, \frac{a}{4}\right)$ be the circumcenter of a triangle with vertices A(a, –2), B(a, 6) and $C\left(\frac{a}{4}, -2\right)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is :

माना A(a, –2), B(a, 6) तथा $C\left(\frac{a}{4}, -2\right)$ के एक त्रिभुज का परिकेन्द्र $\left(5, \frac{a}{4}\right)$ है। यदि इस त्रिभुज के परिवृत्त की त्रिज्या α, त्रिभुज का क्षेत्रफल β तथा त्रिभुज का परिमाप γ है, तो $\alpha + \beta + \gamma$ is बराबर है :

(A) 53 (B) 60 (C) 62 (D) 30

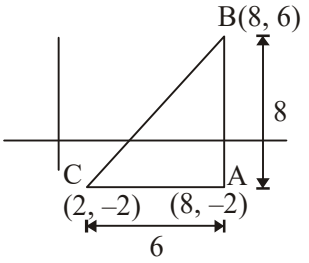
Ans. A

Sol.



Circumcentre lie on perpendicular bisector of AB $\Rightarrow y = 2$

$\Rightarrow \frac{a}{4} = 2$ $\Rightarrow a = 8$



$\Rightarrow$ Perimeter = 6 + 8 + 10 = 24

Area = $\frac{6 \times 8}{2} = 24$

Circumradius = half of hypotenuse

$= \frac{10}{2} = 5$

$\alpha + \beta + \gamma = 24 + 24 + 5 = 53$

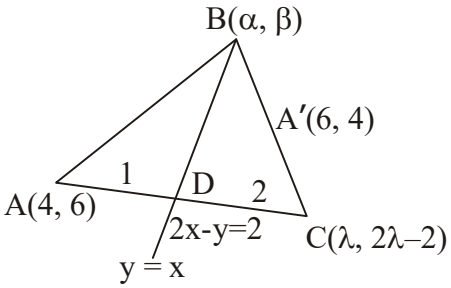
28. In a ΔABC, suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x - y = 2$. If $2AB = BC$ and the points A and B are respectively (4, 6) and (α, β), then $\alpha + 2\beta$ is equal to :

एक ΔABC में माना कोण B के समद्विभाजक का समीकरण $y = x$ है तथा भुजा AC का समीकरण $2x - y = 2$ है। यदि (4, 6) और (α, β) क्रमशः बिन्दु A और B तथा $2AB = BC$ है, तो $\alpha + 2\beta$ बराबर है

(A) 48 (B) 39 (C) 45 (D) 42

Ans. D

Sol.



$\frac{AB}{BC} = \frac{AD}{DC} = \frac{1}{2}$

$D\left(\frac{\lambda+8}{3}, \frac{2\lambda-2+12}{3}\right)$ lies on $y = x$

$$\frac{\lambda+8}{3} = \frac{2\lambda+10}{3}$$

$\Rightarrow \lambda = -2 \Rightarrow c(-2, -6)$

Image of A w.r.t. $y = x$ lies on BC

$m(BC) = m(A'C)$

$$\Rightarrow \frac{\alpha+6}{\alpha+2} = \frac{4+6}{6+2} \Rightarrow \alpha = 14 = \beta$$

$\alpha + 2\beta = 42$

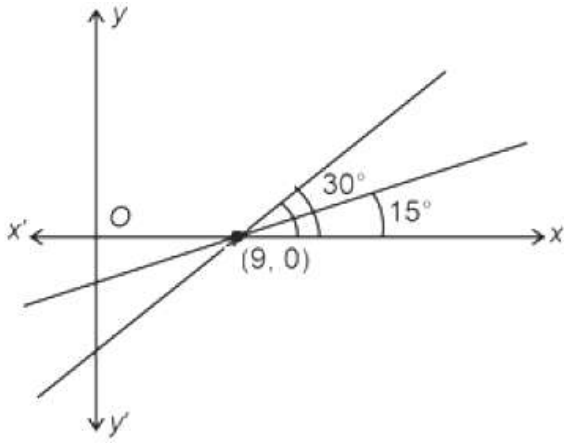
29. A line passing through the point A (9, 0) makes an angle of 30º with the positive direction of x-axis. If this line is rotated about A through an angle of 15º in the clockwise direction, then its equation in the new position is :

बिंदु A (9, 0) से होकर जाने वाली एक रेखा x-अक्ष की धनात्मक दिशा से 30º का कोण बनाती है। यदि इस रेखा को घड़ी की दिशा में 15º के कोण तक A के सापेक्ष घुमाया जाता है, तो नयी स्थिति में इसका समीकरण है :

(A) $\frac{x}{\sqrt{3}-2} + y = 9$ (B) $\frac{x}{\sqrt{3}+2} + y = 9$ (C) $\frac{y}{\sqrt{3}-2} + x = 9$ (D) $\frac{y}{\sqrt{3}+2} + x = 9$

Ans. C

Sol.



Equation : $y - 0 = \tan 15^\circ (x - 9) \Rightarrow y = \left(2 - \sqrt{3}\right)(x - 9)$

30. If $x^2 - y^2 + 2hxy + 2gx + 2fy + c = 0$ is the locus of a point, which moves such that it is always equidistant from the lines $x + 2y + 7 = 0$ and $2x - y + 8 = 0$, then the value of $g + c + h - f$ equals :

यदि एक बिन्दु, जो इस प्रकार चलता है कि इसकी रेखाओं $x + 2y + 7 = 0$ तथा $2x - y + 8 = 0$ से दूरी बराबर रहती है, का बिन्दुपथ

$x^2 - y^2 + 2hxy + 2gx + 2fy + c = 0$ है, तो $g + c + h - f$ का मान बराबर है :

(A) 6 (2) 8 (C) 29 (D) 14

Ans. D

Sol. Locus of point P(x, y) whose distance from

Gives

$X + 2y + 7 = 0$ & $2x - y + 8 = 0$ are equal is $\frac{x+2y+7}{\sqrt{5}} = \pm\frac{2x-y+8}{\sqrt{5}}$

$(x + 2y + 7)^2 - (2x - y + 8)^2 = 0$

Combined equation of lines

$(x - 3y + 1)(3x + y + 15) = 0$

$3x^2 - 3y^2 - 8xy + 18x - 44y + 15 = 0$

$x^2 - y^2 - \frac{8}{3}xy + 6x - \frac{44}{3}y + 5 = 0$

$x^2 - y^2 + 2h\,xy + 2gx + 2fy + c = 0$

$h = \frac{4}{3},\ g = 3,\ f = -\frac{22}{3},\ c = 5$

$g + c + h - f = 3 + 5 - \frac{4}{3} + \frac{22}{3} = 8 + 6 = 14$

31. Let $\alpha, \beta, \gamma, \delta \in Z$ and let $A(\alpha, \beta)$, B(1,0), $C(\gamma, \delta)$ and D(1,2) be the vertices of a parallelogram ABCD. If $AB = \sqrt{10}$ and the points A and C lie on the line $3y = 2x + 1$, then $2(\alpha + \beta + \gamma + \delta)$ is equal to :

माना $\alpha, \beta, \gamma, \delta \in Z$ हैं तथा माना एक समांतर चतुर्भज ABCD के शीर्ष $A(\alpha, \beta)$, B(1,0), $C(\gamma, \delta)$ तथा D(1,2) हैं। यदि $AB = \sqrt{10}$ है तथा बिन्दु A और C रेखा $3y = 2x + 1$ पर हैं, तो $2(\alpha + \beta + \gamma + \delta)$ बराबर है:

(A) 12 (B) 10 (C) 5 (D) 8

Ans. D

Sol. ABCD is a parallelogram so mid-point of AC = mid-point of BD

$$\Rightarrow \left(\frac{\alpha+\gamma}{2}, \frac{\beta+\delta}{2}\right) = \left(\frac{1+1}{2}, \frac{0+2}{2}\right)$$

$$\Rightarrow \left(\frac{\alpha+\gamma}{2}, \frac{\beta+\delta}{2}\right) = (1,1)$$

$\Rightarrow \alpha + \gamma = 2$ and $\beta + \delta = 2$

So, $2(\alpha + \beta + \gamma + \delta) = 2(2 + 2) = 8$

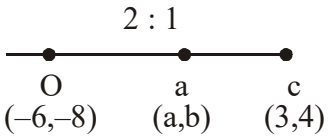
32. Let A(a, b), B(3, 4) and C(–6, –8) respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point P(2a + 3, 7b + 5) from the line $2x + 3y - 4 = 0$ measured parallel to the line $x - 2y - 1 = 0$ is :

माना A(a, b), B(3, 4) तथा C(–6, –8) एक त्रिभुज के केन्द्रक, परिकेन्द्र तथा लंबकेन्द्र है। तो बिन्दु P(2a + 3, 7b + 5) की रेखा $2x + 3y - 4 = 0$ से, रेखा $x - 2y - 1 = 0$ के समांतर नापी गई दूरी है :

(A) $\frac{17\sqrt{5}}{6}$ (B) $\frac{17\sqrt{5}}{7}$ (C) $\frac{\sqrt{5}}{17}$ (D) $\frac{15\sqrt{5}}{7}$

Ans. B

Sol. Centroid divides orthocentre and circumcentre in 2 : 1.

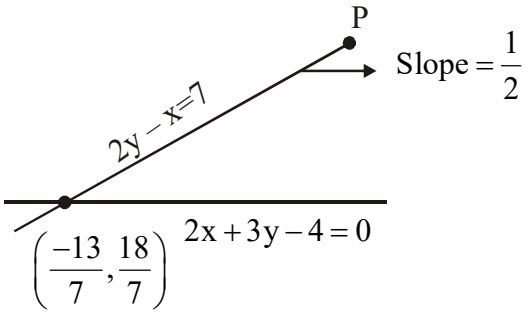


$$a \equiv \frac{2(3)-6}{3} = 0$$

$$b \equiv \frac{2(4)-8}{3} = 0$$

$$\Rightarrow (a,b) \equiv (0,0)$$

$$\Rightarrow P \equiv (3,5)$$



$$\Rightarrow (y-5) = \frac{1}{2}(x-3)$$

$$\Rightarrow 2y - 10 = x - 3$$

$$\Rightarrow 2y - x = 7$$

$$\Rightarrow \sqrt{\left(3+\frac{13}{7}\right)^2 + \left(5-\frac{18}{7}\right)^2} = \frac{\sqrt{34^2+17^2}}{7}$$

$$= \frac{17\sqrt{5}}{7}$$

33. If the sum of squares of all real values of α, for which the lines $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is p, then the greatest integer less than or equal to p is _____.

माना α के सभी वास्तविक मानों, जिनके लिए रेखाएँ $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ तथा $\alpha x + 2y - 2 = 0$ एक त्रिभुज नहीं बनाती हैं, के वर्गों का योग p है, तो महत्तम पूर्णांक $\le p$ है _____

Ans. 32

Sol. $2x - y + 3 = 0$

$6x + 3y + 1 = 0$

$\alpha x + 2y - 2 = 0$

will not form a Δ if $\alpha x + 2y - 2 = 0$ is concurrent with $2x - y + 3 = 0$ and $6x + 3y + 1 = 0$ or parallel to either of them so

Case-1: Concurrent lines

$$\begin{vmatrix} 2 & -1 & 3 \\ 6 & 3 & 1 \\ \alpha & 2 & -2 \end{vmatrix} = 0 \Rightarrow \alpha = \frac{4}{5}$$

Case-2 : Parallel lines

$$-\frac{\alpha}{2} = \frac{-6}{3} \text{ or } -\frac{\alpha}{2} = 2$$

$\Rightarrow \alpha = 4$ or $\Rightarrow \alpha = -4$

$$P = 16 + 16 + \frac{16}{25}$$

$$[P] = \left[32 + \frac{16}{25}\right] = 32$$

34. Let A(6, 8), B(10 cos α, – 10 sin α) and C(–10 sin α, 10 cos α), be the vertices of a triangle. If L(a, 9) and G(h, k) be its orthocenter and centroid respectively, then $(5a - 3h + 6k + 100 \sin 2\alpha)$ is equal to _____.

माना कि A(6, 8), B(10 cos α, – 10 sin α) और C(–10 sin α, 10 cos α) एक त्रिभुज के शीर्ष हैं। यदि L(a, 9) और G(h, k) इसके क्रमशः लम्बकेंद्र और केन्द्रक हैं, तो $(5a - 3h + 6k + 100 \sin 2\alpha)$ बराबर है _____।

Ans. 145

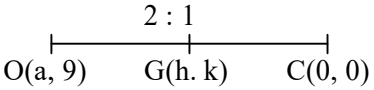
Sol. A(6, 8)

B(10 cos θ, – 10 sin α)

C(–10 sin α, 10 cos α)

Lies on circle $x^2 + y^2 = 10^2$

so circumcentre (C) = (0, 0)



$\frac{O+a}{3}=h$ | $\frac{9+0}{3}=k$

$a=3h$ | $k=3$

Now $h=\frac{6+10\cos\alpha-10\sin\alpha}{3}$ $\quad k=3=\frac{8-10\sin\alpha+10\cos\alpha}{3}3$

$3h = 6\ 10\cos\alpha - 10\sin\alpha$ $\quad 10\cos\alpha - \sin\alpha = 1$

$3h = 6 + 1$

$h = 7/3 \quad 3h = 7 \quad a = 7$

$1 = 10(\cos\alpha - \sin\alpha)$

square

$1 = 100(1 - \sin 2\alpha)$

$100\sin 2\alpha = 99$

Now $5a - 3h + 6k + 100\sin 2\alpha$

$\Rightarrow 35 - 7 + 18 + 99$

$= 145$

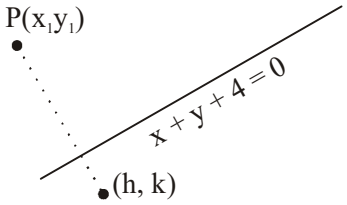
35. Let A and B be the two points of intersection of the line $y + 5 = 0$ and the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If d denotes the distance between A and B, and a denotes the area of ΔSAB, where S is the focus of the parabola $y^2 = 4x$, then the value of $(a + d)$ is _____.

माना कि परवलय $y^2 = 4x$ के रेखा $x + y + 4 = 0$ में दर्पण प्रतिबिम्ब और रेखा $y + 5 = 0$ के प्रतिच्छेद बिन्दु A और B हैं। यदि A और B हैं। यदि A और B के बीच की दूरी d है और ΔSAB का क्षेत्रफल a है, जहाँ S परवलय $y^2 = 4x$ की नाभि है, तो $(a + d)$ का मान है _____

Ans. 14

Sol. Mirror image of $y^2 = 4x$

with respect to $x + y + 4 = 0$



$$\Rightarrow\left(\frac{h-x_1}{1}\right)=\left(\frac{k-y_1}{1}\right)=-\frac{2\left(x_1+y_1+4\right)}{1^2+1^2}$$

$\Rightarrow h - x_1 = k - y_1 = -x_1 - y_1 - 4$

$h = -y_1 - 4 \Rightarrow -(h + 4) = y_1$

$k = -x_1 - 4 \Rightarrow -(k + 4) = x_1$

$y_1^2 = 4x_1$

$(-(h + 4))^2 = -4k - 16$

$\Rightarrow h^2 + 8h + 4k + 32 = 0$

$\Rightarrow x^2 + 8x + 4y + 32 = 0$ (Equation of mirror image of $y^2 = 4x$ w.r.t. $x + y + 4 = 0$)

Intersection with $(y + 5 = 0)$

$y = -5$

$x^2 + 8x + 4(-5) + 32 = 0$

$x^2 + 8x + 12 = 0$

$(x + 6)(x + 2) = 0$

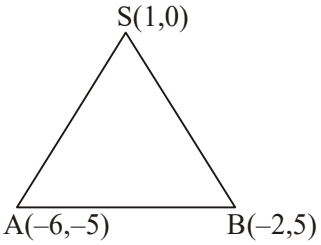
$x = -6$ & $x = -2$

$(-6, -5)$	$(-2, -5)$	$(1, 0)$
Point A	Point B	Point S (Focus of $y^2 = 4x$)

d = Distance between A & B

d = 4

a = ar(ΔSAB)



$$a = \frac{1}{2}\begin{vmatrix} 1 & 0 & 1 \\ -6 & -5 & 1 \\ -2 & -5 & 1 \end{vmatrix}$$

$R_2 \to R_2 - R_3$

$$a = \frac{1}{2}\begin{vmatrix} 1 & 0 & 1 \\ -4 & 0 & 0 \\ -2 & -5 & 1 \end{vmatrix} = \left(\frac{1}{2}\right)(4)(5)$$

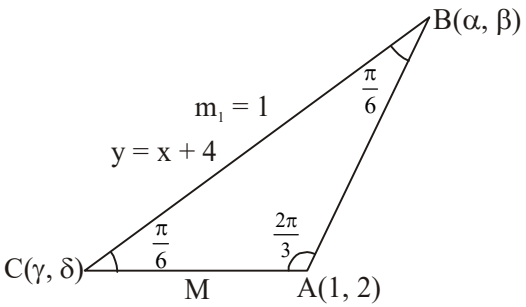
$\Rightarrow (a + d) = 10 + 4 = 14$

36. Consider a triangle ABC having the vertices A(1,2), B(α, β) and C(γ, δ) and angles $\angle ABC = \frac{\pi}{6}$ and $\angle BAC = \frac{2\pi}{3}$. If the points B and C lie on the line $y = x + 4$, then $\alpha^2 + \gamma^2$ is equal to _____.

एक त्रिभुज ABC जिसके शीर्ष A(1,2), B(α, β) तथा C(γ, δ) और कोण $\angle ABC = \frac{\pi}{6}$ तथा $\angle BAC = \frac{2\pi}{3}$ हैं पर विचार कीजिए। यदि बिन्दु B तथा C रेखा $y = x + 4$ पर स्थित हैं, तो $\alpha^2 + \gamma^2$ बराबर है _____

Ans. 14

Sol.



Equation of line passing through P(1, 2) and making angle $\frac{\pi}{6}$ with $y = x + 4$ _____(1)

$y - 2 = m(x - 1)$

$y - 2 = \left(2 \pm \sqrt{3}\right)(x - 1)$ _____(2)

$\tan\frac{\pi}{6} = \left|\frac{m - m_1}{1 + mm_1}\right| \pm \frac{1}{\sqrt{3}} = \frac{m-1}{1+m}$

$m = 2 \pm \sqrt{3}$

Solve (1) and (2)

$$x + 4 - 2 = \left(2 \pm \sqrt{3}\right)(x - 1)$$

$$x = \frac{4+\sqrt{3}}{1+\sqrt{3}} = \alpha \text{ and } x = \frac{4-\sqrt{3}}{1-\sqrt{3}} = \gamma$$

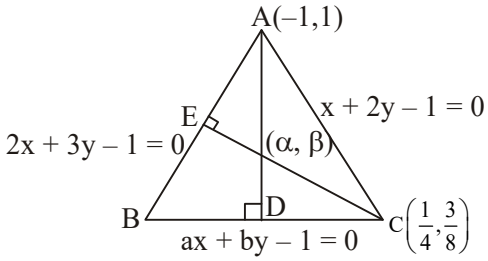
$\therefore \alpha^2 + \gamma^2 = 14$

37. If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are (3, 4) and (–6, –8), then the value of $|a - b|$ is _____.

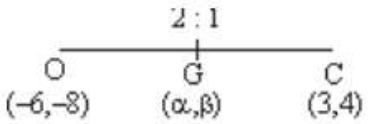
यदि रेखाओं $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ और $ax + by - 1 = 0$ द्वारा बनाया गया त्रिभुज, जिसका लंब केन्द्र किसी अन्य त्रिभुज के केन्द्रक के बराबर है एवं जिसका परिकेन्द्र और लंब केन्द्र क्रमशः (3, 4) तथा (–6, –8) है, तो $|a - b|$ का मान होगा _____

Ans. 16

Sol.



For another Δ



$$\alpha = \frac{2\times 3-6}{3} = 0 \qquad \beta = \frac{2\times 4-8}{3} = 0$$

$$(\alpha,\beta) = (0,0)$$

$$x + 2y - 1 = 0$$
$$2x + 3y - 1 = 0$$
$$\underline{- \; - \quad +}$$
$$y - 1 = 0$$

$$y = 1$$
$$x = -1$$

A(–1, 1)

$$\text{Slope of AD} = \frac{0-1}{0+1} = -1$$

and

$$M_{AD} \times M_{BC} = -1$$
$$(-1)\, M_{BC} = -1$$
$$M_{BC} = 1$$

Now for point C

find equation of CE

$$M_{CE} = \frac{2}{3}$$

$$y = \frac{2}{3}x$$

and equation of AC

$$x + 2y - 1 = 0$$
$$\underline{-3x + 2y = 0}$$
$$4x - 1 = 0$$

$$x = \frac{1}{4}$$

$$y = \frac{3}{2}\times\frac{1}{4} = \frac{3}{8}$$

Eq. of BC

$C\left(\frac{1}{4},\frac{3}{8}\right)$

$m_{BC} = 1$

$y-\frac{3}{8}=1\left(x-\frac{1}{4}\right)$

$\frac{8y-3}{8}=\frac{4x-1}{4}$

$8y - 8x + 2 - 3 = 0$

$-8x + 8y - 1 = 0$

$ax + by - 1 = 0$

$a = -8$

$b = 8$

Value of $|a - b| = |-8 - 8| = 16$